

Thought-to-Text with EEG — Deep Dive & Roadmap (v1.0)

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Team: Thought-to-Text (T2T)

Scope: A complete technical deep dive and actionable plan to build an EEG-based “thought-to-text” system, including theory, data, modeling options, evaluation, roadmap, and a curated reading list.

Target: Transition from small-vocabulary imagined-speech classification toward open-vocabulary decoding with reliable user-level adaptation.

1) Executive Summary

- **Problem framing.** We aim to convert non-invasive EEG activity during *inner/imagined speech* into text. Because EEG is low-SNR and spatially coarse, we should treat decoding as *noisy sequence transduction* (neural activity → subword sequence), assisted by strong priors (language models) and by per-user adaptation.
- **Near-term goal (3–4 months).** Robust, real-time **keyword/phrase** decoding for a *subject-dependent* setting using ~10–20 frontal-temporal electrodes, with subword CTC decoding and n-gram LM fusion.
- **Mid-term goal (6–9 months).** **Open-vocabulary** decoding in constrained domains (e.g., phrasebooks or task commands), **few-shot subject adaptation**, and **cross-subject pretraining** via self-supervised learning (SSL).
- **Long-term goal (12–18 months).** Multi-day on-device model with **continual calibration**, **active privacy controls**, and **LLM-assisted error correction**; explore multi-modal add-ons (EOG/EMG) for artifact suppression and articulatory hints.

High-level recipe: 1. **Clean signals well** (artifact subspace reconstruction + notch/band-pass + channel QA).
2. **Good front-end** (EEGNet/Conv front-end → Transformer/Conformer encoder).
3. **Sequence loss** (CTC over characters/phonemes; optional RNN-T for streaming).
4. **Language priors** (CTC beam search + wordpiece LM; LLM re-scoring in batch mode).
5. **Subject adaptation** (LayerNorm/BatchNorm adaptation + LoRA fine-tuning + few minutes of calibration).
6. **SSL pretraining** on mixed EEG corpora (contrastive/MAE-style) for cross-subject robustness.

2) Background & Theory

2.1 Inner vs. Imagined Speech and Why EEG is Hard

- **Inner speech** (silent verbal thought) and **imagined speech** (silently producing words) share circuits with overt speech but produce weaker, more variable signals.
- EEG provides **millisecond temporal resolution** but limited spatial specificity; scalp sensors receive mixtures of cortical sources and **muscle artifacts** (facial EMG, micro-movements).

- Practical implication: models should emphasize **temporal dynamics**, **cross-frequency patterns**, and **strong priors** over the output sequence.

2.2 Where the Signal Lives (neuro hints)

- Frontal and temporal networks (IFG/Broca's area, STG/MTG), sensorimotor cortices, and parietal hubs contribute to speech imagery. For non-invasive EEG, **fronto-temporal and central** montages are typically most informative.

2.3 Spectral-Temporal Structure

- Relevant bands often include **theta/alpha** (attention/linguistic integration), **beta** (motor planning) and sometimes **low-gamma**; cross-frequency coupling can encode phonetic categories.

2.4 Problem Formulation

- Let $x_{1:T}$ be multi-channel EEG; we seek $y_{1:N}$ (characters or wordpieces). Alignment is unknown; therefore use **CTC** (or RNN-T) with **blank** symbols and **beam search**.

3) Data Landscape (for training, benchmarking, and pretraining)

Inner/Imagined Speech & Speech-Related EEG: - **Imagined-Speech corpora (recent)**. Small- to medium-scale datasets with words/syllables, often subject-dependent sessions. Include newer releases focused on imagined vowels/words and cueless paradigms. - **Reading EEG corpora (ZuCo 1.0/2.0, DERCo)**. EEG aligned to text during reading; excellent for **EEG↔language** representation learning (EEG-to-text pretraining, EEG-conditioned language tasks). - **Large-hours single-subject speech EEG** collections demonstrate **scaling effects** (performance grows sharply with hours).

Recommendation: For open-vocabulary ambitions, **mix reading-EEG (text-aligned)** with **imagined-speech EEG**, then pretrain with SSL and fine-tune on the imagined-speech objective.

4) Signal Processing & Pre-Processing Pipeline

1. **Channel QA & Referencing**
2. Drop flat/noisy channels; common average reference (CAR) or Laplacian when appropriate.
3. **Filtering**
4. **Band-pass** (e.g., 0.5–45 Hz) + **notch** at mains frequency; verify transition bandwidths and avoid filter ringing.
5. **Artifact removal**
6. **ASR (Artifact Subspace Reconstruction)** to repair high-variance bursts; optional ICA for ocular/muscle components.
7. Log **EOG** (vertical/horizontal) and optional **facial EMG** to help regress residual artifacts and to *detect/ignore* contaminated windows.
8. **Segmentation**

9. Sliding windows (0.5–2.0 s) with overlap (25–75%) for streaming; align windows to cues when available.

10. **Normalization**

11. Per-channel robust scaling; **subject-wise** normalization statistics cached for adaptation.

Deliverable: A reproducible **MNE/EEGLAB** script (“prep.py”) that standardizes the above with configurable params and audit logs.

5) Feature Representations

- **End-to-end raw** (Conv front-ends learn temporal and spatial filters).
- **Time-frequency** (STFT or wavelet scalograms; supports SpecAugment-style masking).
- **Riemannian covariance** (xDAWN / SPD covariances → **tangent-space** features) for strong subject-robust baselines and model ensembles.
- **Graph features** (functional connectivity graphs → GNN encoders) as auxiliary paths.

Recommendation: Start with **EEGNet**/ShallowConvNet features feeding a **Conformer** encoder; add a parallel **Riemannian tangent-space branch** whose logits are fused via learnable gating.

6) Model Architectures

6.1 Baseline (what we have vs. what we propose)

- **Current (in repo):** LSTM sequence model trained per subject.
- **Proposed v0.1 (weeks 1–4):**
- **Front-end:** EEGNet (depthwise + separable convs) or ShallowConvNet.
- **Encoder:** 6–12-layer **Transformer/Conformer** with relative position encodings; causal masks for streaming.
- **Output:** CTC over **characters or 100–500 wordpieces** (BPE).
- **Decoder:** CTC beam search with **n-gram LM** (KenLM/pyctcdecode).

6.2 Mid-term upgrades (v0.2–v0.3)

- **RNN-T** (encoder + predictor + joiner) for lower-latency streaming.
- **Multi-task heads:** (i) keyword spotting, (ii) phoneme inventory, (iii) wordpiece CTC — encourages disentangled representations.
- **SpecAugment-style** masks on time-frequency inputs; **cutout/mixup** on channels.
- **SSL pretraining:** TS2Vec / masked autoencoder for EEG (MAE-EEG) on mixed corpora; linear-probe sanity checks.
- **Domain adaptation:** AdaBN/BatchNorm re-estimation per subject; **CORAL/DANN** alignment across subjects.
- **Ensembling:** Late fusion between Conv-CTC and Riemannian-LogReg; Bayesian model averaging for confidence.

6.3 Long-term (v1.x)

- **Cross-modal** pretraining with text/phonetics (align EEG embeddings to phoneme/wordpiece embeddings).
 - **LLM re-scoring** of n-best hypotheses using shallow fusion constraints; instruction-tuned LLM for **error-aware correction** (e.g., homophones, phonotactics).
 - **Personalization**: LoRA adapters per user; *few-shot* fine-tune using 3–10 minutes of their data → measurable drop in CER/WER.
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7) Training, Evaluation & Metrics

7.1 Splits & Protocols

- **Within-subject** (easy) vs. **cross-subject** (hard). Report both.
- For online viability: run **leave-session-out** and **leave-day-out** splits.

7.2 Metrics

- **Keyword**: accuracy, F1, AUROC.
- **Sequence**: **CER/WER** (character/word error rate) using CTC decoding; report **latency** (ms) and **real-time factor (RTF)**.
- **Calibration cost**: time needed to reach target CER.

7.3 Ablations (must-do)

- Window size/overlap; front-end choice (EEGNet vs. ShallowConvNet); CTC vocab size; beam width and LM weight; normalization/adaptation strategies; effect of SSL pretraining; artifact-handling (ASR on/off); electrode subsets.

7.4 Statistical Hygiene

- **Permutation tests** against chance; confidence intervals via subject-level bootstraps; preregistered metrics to avoid p-hacking.
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8) Engineering Plan & Roadmap

Stage A — Foundations (Weeks 1–4) - Implement standardized **prep.py** with ASR + filter + QA.

- Reproduce **EEGNet + Conformer + CTC** baseline; greedy and beam decoding; log CER/WER.
- Integrate **KenLM/pyctcdecode** for LM fusion; add a CLI for beam width, LM weight, pruning.
- Deliver **real-time demo** (keyboard control + few keywords) with <250 ms end-to-end latency on a 1-s context.

Stage B — Robustness (Weeks 5–10) - **SSL pretraining** on reading EEG (ZuCo/DERCo) + imagined-speech corpora; evaluate linear probe; fine-tune on target task.

- Add **subject adaptation** (AdaBN + small LoRA on top-k layers).

- Introduce **Riemannian branch** and late-fusion; run cross-subject benchmarks.
- Ship **calibration wizard** (5–10 min script with active learning of hard tokens/words).

Stage C — Open-Vocabulary (Weeks 11–16) - Expand vocab to **wordpieces** (e.g., 256–512 BPE).

- Add **phoneme auxiliary loss** (CTC multi-task); evaluate on command-set and short phrases.
- Integrate **LLM n-best rescoring** (offline) and **error-aware correction** (confusion patterns).

Stage D — Productization (Weeks 17–24) - Optimize **runtime** (vectorized beam search, GPU decoder); RTF < 0.2 on laptop GPU.

- Build **privacy controls** (user on/off gating, delete-my-data, local-only inference mode).
- Documentation + evaluation cards; human-factors study for usability/mental workload.

Success gates: - A: Keyword demo, subject-dependent CER<40% (short phrases).

- B: Cross-subject keyword F1>0.75; adaptation reduces CER by $\geq 20\%$ relative.

- C: Open-vocab phrases CER<55% with LM fusion; <400 ms latency.

- D: Stable 30-min sessions without drift; user-visible privacy controls.

9) Practical Modeling Details

- **Vocabulary units:** Start with **characters**; quickly move to **wordpieces** (~256–512) to reduce insertion/deletion errors.
- **Beam search:** width 25–50; tunable LM weight and insertion bonus; export n-best lists.
- **Augmentations:** random channel dropout, Gaussian jitter, time-warping; SpecAugment on TF inputs (time/frequency masking).
- **Regularization:** stochastic depth on encoder; weight decay; label smoothing.
- **Calibration:** fast BN re-estimation + 1–2 epochs LoRA on last 2–3 blocks.
- **Confidence:** entropy/CTC posterior collapse detectors to trigger *ask-to-repeat* prompts.

10) Risk Register & Mitigations

Risk	Impact	Mitigation
EMG/EOG contamination (micro-movements)	False positives; illusory decoding	EOG/EMG channels; ASR + ICA; window rejection; bite-bar or jaw-rest for tests; video capture for audits.
Dataset mismatch (overt vs. imagined vs. reading)	Poor generalization	Multi-task training; curriculum (overt→imagined); use reading EEG for SSL only; domain adaptation (AdaBN/DANN/CORAL).
Cross-subject variability	Unstable performance	Subject-wise normalization; few-shot adaptation; Riemannian features; large-subject SSL.

Risk	Impact	Mitigation
Latency	Not real-time	Causal encoders; RNN-T; GPU-vectorized CTC decoding.
Privacy & consent	Trust, compliance	Local inference option; explicit user-armed capture; data minimization; transparent logs; consent UX.

11) What “Good” Looks Like (KPIs)

- **Subject-dependent** short-phrase CER < 35–40% with LM fusion.
- **Cross-subject** keyword F1 > 0.80 after ≤ 10 min calibration.
- **Latency** < 300–400 ms for 1-s look-back.
- **Stability**: No >10% CER drift across 30-min sessions.

12) Implementation Checklist (first pass)

- [] `prep.py` (ASR+filter+QA)
- [] `train_ctc.py` (EEGNet/Conformer)
- [] `decode.py` (greedy/beam + KenLM)
- [] `ssl_pretrain.py` (TS2Vec / MAE-EEG)
- [] `adapt.py` (AdaBN + LoRA)
- [] `eval.py` (CER/WER, latency, ablations)
- [] `demo/` (real-time GUI + privacy toggle)

13) Reading List (prioritized)

A. Core Methods & Architectures

1. EEGNet; Shallow/Deep ConvNets for EEG; Conformer encoders; CTC sequence modeling; vectorized/efficient CTC decoding; pyctcdecode/KenLM usage.
2. Riemannian EEG (xDawn, SPD covariance, tangent-space).
3. SSL for EEG & time-series (TS2Vec, MAE for EEG, contrastive objectives).
4. Domain adaptation: AdaBN, CORAL, DANN, tangent-space alignment.

B. Imagined/Inner Speech with EEG

1. Systematic reviews; state-of-the-art surveys.
2. Recent datasets (imagined vowels/words; cueless imagined speech).
3. Cross-modal links (reading EEG $\leftrightarrow$ text; listened speech alignment).
4. Evidence for frontal-temporal involvement; low-frequency/cross-frequency cues.

C. Ethics & Safety

1. Mental privacy and neurorights; user-control/agency in speech-BCIs; practical governance.

A fully annotated bibliography with links appears in **Section 15** below.

14) “Interesting Info” (fast facts for the team)

- **Scaling matters:** moving from ~10 hours to 175 hours of EEG for a single subject can raise top-1 recognition from ~2–3% to ~50% in zero-shot segment tasks.
 - **Low-freq power & cross-frequency coupling** carry substantial imagined-speech information—even when high-gamma is missing (EEG can still exploit low bands).
 - **Cueless paradigms** may better approximate natural inner speech but are harder; accuracy improves with *overt*→*imagined* curriculum and subject-specific calibration.
 - **Riemannian baselines** remain strong and are great for small-data subjects and sanity checks.
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15) References (APA style with links)

(A curated subset most relevant to our plan; grouped by theme)

15.1 Methods & Tooling

- Graves, A., Fernández, S., Gomez, F., & Schmidhuber, J. (2006). Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. *ICML*. https://www.cs.toronto.edu/~graves/icml_2006.pdf
- Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018). EEGNet: A compact convolutional neural network for EEG-based BCIs. *Journal of Neural Engineering*, 15(5), 056013. <https://arxiv.org/abs/1611.08024>
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- MNE-Python docs. (2025). Filtering background. https://mne.tools/stable/auto_tutorials/preprocessing/25_background_filtering.html
- EEGLAB **clean_rawdata** (ASR) plugin. (n.d.). https://eeglab.org/plugins/clean_rawdata/

15.2 Riemannian & Alignment

- Barachant, A., Bonnet, S., Congedo, M., & Jutten, C. (2013). Classification of covariance matrices using Riemannian geometry. *IEEE TBME*, 59(4), 920–928. (Classic reference; see survey implementations).
- Lai-Tan, N., Gu, X., Philiastides, M. G., & Deligianni, F. (2025). Cross-subject transfer via individual tangent-space alignment. *arXiv*. <https://arxiv.org/abs/2508.08216>

15.3 Self-Supervised Learning (EEG & Time-series)

- Yue, Z., et al. (2021). TS2Vec: Universal representation of time series. *arXiv*. <https://arxiv.org/abs/2106.10466>
- Yang, C., et al. (2023). Self-supervised EEG representations (ContraWR). *JMIR AI*. <https://ai.jmir.org/2023/1/e46769>
- Zhang, M., et al. (2024). SSL for EEG-based emotion recognition. *Frontiers in Human Neuroscience*. <https://www.frontiersin.org/articles/10.3389/fnhum.2024.1334721/full>
- Qiao, Y., et al. (2025). Self-supervised data augmentation for EEG. *Applied Soft Computing*. <https://www.sciencedirect.com/science/article/abs/pii/S1566253525003525>

15.4 Datasets & Empirical Benchmarks

- Zhang, Z., et al. (2024). An EEG imagined-speech BCI dataset. *Scientific Data*. <https://www.nature.com/articles/s41597-024-04114-1>
- Hollenstein, N., et al. (2018). ZuCo: EEG + eye-tracking during reading. *Scientific Data*, 5, 180291. <https://www.nature.com/articles/sdata2018291>
- Hollenstein, N., et al. (2020). ZuCo 2.0. *LREC*. <https://aclanthology.org/2020.lrec-1.18.pdf>
- Quach, B. M., et al. (2024). DERCo: Reading EEG for NLP. *Scientific Reports*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11464549/>
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- Derakhshesh, A., et al. (2025). Cueless imagined-speech dataset. *arXiv*. <https://arxiv.org/abs/2501.09700>

15.5 Inner/Imagined Speech Evidence

- Lopez-Bernal, D., et al. (2022). EEG imagined speech SoTA review. *Frontiers in Neuroscience*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9086783/>
- Proix, T., et al. (2022). Imagined speech from low- and cross-frequency signals (ECoG). *Nature Communications*, 13, 48. <https://www.nature.com/articles/s41467-021-27725-3>
- Wandelt, S. K., et al. (2024). Internal speech at single-neuron level (SMG). *Nature Human Behaviour*. <https://www.nature.com/articles/s41562-024-01867-y>
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15.6 Artifacts & Preprocessing

- Miyakoshi, M. (2023). Artifact subspace reconstruction overview. *Frontiers in Human Neuroscience*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10710985/>
- EEGLAB **clean_rawdata** docs & tutorial. https://eeglab.org/tutorials/06_RejectArtifacts/cleanrawdata.html
- Porcaro, C., et al. (2015). Removing speech artifacts; EMG issues. *NeuroImage*. <https://pubmed.ncbi.nlm.nih.gov/25450111/>
- Delorme, A., et al. (2023). EEG is better left alone (preprocessing cautions). *eLife*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9911389/>

15.7 Ethics & Governance (mental privacy/user control)

- Yang, H., et al. (2025). Regulating neural data processing in the age of BCIs. *Nature Human Behaviour*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11951885/>
- van Stuijvenberg, O. C., et al. (2024). User control in speech-BCIs. *Frontiers in Human Neuroscience*. <https://www.frontiersin.org/articles/10.3389/fnhum.2024.1420334/full>
- Brown, C. M. L. (2024). Neurorights & mental privacy. *Neuroethics*. <https://link.springer.com/article/10.1007/s12152-024-09568-z>

(Links were verified as of Sept 16, 2025; many are Open Access.)

16) Appendix: Concrete Spec (v0.1 Baseline)

Input: 16–32 channels (10–20 montage concentrating frontal/temporal/central), 250–512 Hz sampling.

Preproc: 0.5–45 Hz FIR band-pass; notch at 50/60 Hz; ASR repair; ICA ocular component removal; per-channel robust z-score; 1.0-s windows, 0.5-s hop.

Model: EEGNet (F1=8 temporal filters; depthwise spatial conv across channels; separable conv); Conformer (d=256, L=8, heads=4; conv kernel=15; dropout=0.1); CTC over 256 BPE tokens + blank.

Decode: Beam=25; LM=4-gram KenLM; LM weight=0.6; insertion bonus=0.2.

Train: AdamW (lr=3e-4), cosine decay, wd=0.05, 80 epochs; SpecAugment (time mask up to 80 ms; freq mask up to 3 bands); channel dropout 0.1.

Eval: CER/WER on held-out session; latency & RTF; calibration ablation (AdaBN only vs. AdaBN+LoRA).

Artifacts: Reject windows with EOG RMS> τ or EMG band power> τ ; log rejection rate.

17) Appendix: Subject Adaptation (v0.2)

- **AdaBN:** recompute BN stats on 2–5 minutes of subject data.
 - **LoRA:** rank-8 adapters on top-2 encoder blocks (K,V projections).
 - **Prompted calibration:** pick hard phonemes/wordpieces via uncertainty sampling; few extra minutes for gains.
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18) Appendix: Governance & UX Patterns

- **On-device switch:** decoding only when the user arms the system (hotkey/gesture).
- **“Delete my session”** & “export my data” buttons; transparent session logs.
- **Privacy banner:** we decode *only when armed*; no background capture.
- **Red-team tests:** deliberate artifact injection (jaw clench, eye saccades) to verify robustness.